Investigating the Latent Roots of Distraction in Retrieval-Augmented Generation via Sparse Autoencoder Interventions

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Abstract

Large language models exhibit a critical vulnerability to distractor interference in retrieval-augmented contexts: they fail to prioritize relevant, factually correct documents over topically similar but misleading content. We introduce Lat-Defuse, a mechanistic framework that corrects this failure mode through targeted interventions in the model's latent space. Using Sparse Autoencoders (SAEs), our method operates in an interpretable feature space and formulates correction as constrained counterfactual optimization. On Gemma-2 and Llama-3 model families across three QA benchmarks (BioASQ, Natural Questions, PopQA), our method achieves recovery rates of up to 94% on distractor-vulnerable samples. Successful correction through sparse modifications reveals distractor interference as a localized, systematically addressable phenomenon, opening directions toward universal distractor robustness in LLMs.

1. Introduction

Large language models exhibit a critical vulnerability when processing (retrieval-augmented) contexts: they struggle to distinguish between relevant, factually correct information, and topically similar but misleading content [2, 3, 4]. This weakness manifests prominently in two related phenomena: the "lost-in-the-middle" effect in retrieval-augmented generation (RAG) systems, where models fail to prioritize gold documents surrounded by distractors [5, 6, 7], and context rot [8] in long-context scenarios, where models lose focus amid extensive but partially irrelevant information.

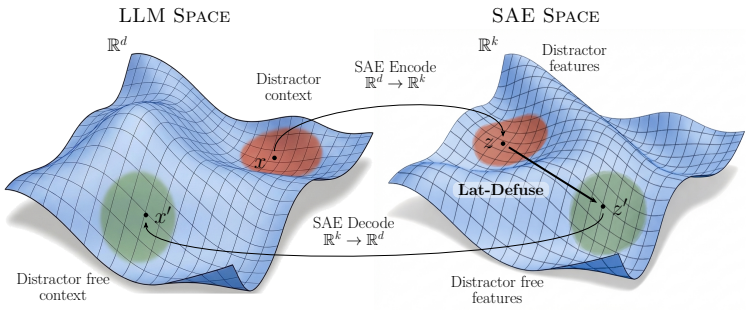


Figure 1: Lat-Defuse overview. Distractor-corrupted activations $x \in \mathbb{R}^d$ are encoded into SAE feature space, yielding latent representation $z \in \mathbb{R}^k$. Constrained optimization produces z' by modulating distractor-associated features while preserving gold-document cues. Decoding yields corrected activations x', restoring factual generation.

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Current approaches to mitigating distractor interference operate at two distinct levels. Input-side methods [9] focus on improving retrieval quality through enhanced document ranking and filtering mechanisms, while reasoning-side approaches [10] aim to improve the model's ability to critically evaluate and synthesize information through chain-of-thought prompting or verification procedures. However, these methods incur computational overhead and function as architectural stopgaps that address symptoms without investigating the underlying cause. A central question remains unanswered: **which representational changes occur in the model's latent space when topically similar but factually incorrect documents induce generation errors?** Without understanding how distractors corrupt internal representations, solutions remain limited to symptom-level mitigation.

To investigate whether distractor interference can be defused in latent space, we study the maximally challenging configuration: contexts containing the gold document alongside hard distractors that induce generation errors. Scenarios involving irrelevant noise or weaker distractors represent relaxations of this setting. We introduce Lat-Defuse, a mechanistic framework for analyzing and correcting distractor interference through direct intervention in the model's latent space. Using Sparse Autoencoders (SAEs) to operate in an interpretable feature space, our method formulates correction as constrained optimization targeting the point at which generation first departs from factual correctness, identifying the minimal representational changes required to restore factual accuracy.

We evaluate our approach on Gemma-2 and Llama-3 model families across three QA benchmarks (BioASQ, Natural Questions, PopQA), achieving recovery rates up to 94% on distractor-vulnerable samples. Successful correction through sparse feature modifications indicates that distractor interference operates through **localized** representational changes rather than global perturbations. Our contributions are: **(a)** we introduce SAE-based latent intervention as a mechanistic framework for characterizing how distractor interference degrades factual accuracy in RAG; **(b)** we demonstrate that distractor interference can be corrected through targeted modifications to a sparse set of interpretable features; **(c)** we show that the correction mechanism generalizes across architectures (Gemma-2, Llama-3) and model scales. This mechanistic characterization provides a foundation toward universal solutions for distractor robustness in language models. [1]

2. Methodology

2.1. Latent-Space Intervention

Let M be a language model over vocabulary $\mathscr{V}$. We consider a retrieval-augmented setting with input query q and context $C = \{d_g, d_{\text{dist}}\}$, comprising a gold document d_g and semantic distractors d_{dist}. The model's predictions deviate from the ground-truth sequence Y^*. We target the first divergence point: the earliest token index t where the model's argmax prediction differs from Y_t^*. Let $y^* \in \mathscr{V}$ denote the correct token at this position. We intervene on the residual stream of M via a pre-trained Sparse Autoencoder (SAE) with encoder $E : \mathbb{R}^d \to \mathbb{R}^K$ and decoder $D : \mathbb{R}^K \to \mathbb{R}^d$. We define an intervention window over the final W tokens of the input context. For each token index i, the *modulated activations* $\tilde{a}_i = D(z_i)$ are the decoder output for latent state $z_i \in \mathbb{R}^K_{\geq 0}$, where non-negativity is enforced to match the SAE's ReLU gating. The optimization objective $\mathscr{L}$ is the cross-entropy loss at divergence position t, conditioned on these modulated activations:

$$\mathscr{L} = -\log P_M(y^* \mid q, C, \tilde{a}) \tag{1}$$

Optimization terminates when the prediction correction condition holds:

$$\arg\max_{v \in \mathscr{V}} P_M(v \mid q, C, \tilde{a}) = y^* \tag{2}$$

We optimize the latent features with projected gradient updates, initializing $z_i^{(0)} = E(a_i^{(0)})$ from the original forward-pass activations $a_i^{(0)}$ to start from the distractor-corrupted state. The parameters of

[1] Official code and instructions to reproduce the experiments are available at https://github.com/christian-giannetti/mechanics-of-interference.

M and the SAE remain frozen; only z_i is updated. Each iteration k follows a three-step cycle: (i) decode the latent state to produce modulated activations $\tilde{a}_i = D(z_i^{(k)})$, (ii) inject $\tilde{a}_i$ into the computation graph of M to compute $\nabla_{z_i}\mathscr{L}$, and (iii) apply a projected gradient step enforcing non-negativity via ReLU. The update rule is:

$$z_i^{(k+1)} \leftarrow \text{ReLU}\left(z_i^{(k)} - \eta\nabla_{z_i}\mathscr{L}(z_i^{(k)})\right) \tag{3}$$

where η is the learning rate. This procedure yields a counterfactual latent state z^* satisfying the prediction correction condition. The feature-space delta $\Delta z = z^* - z^{(0)}$ identifies the modification sufficient to reverse the distractor-induced error.

3. Experimental Setup

3.1. Models and Sparse Autoencoders

We evaluate our approach using the Gemma-2 (2B) and Llama-3.1 (8B) model families.[2] To facilitate latent space interventions, we use Gemma Scope [11] and Llama Scope [12] Sparse Autoencoders (SAEs) with dictionary sizes $K \in \{16\text{k}, 32\text{k}\}$.[3] We intervene at four network depths to assess layer-wise susceptibility: embedding (0%), early-mid (40%), mid-late (70%), and late (90%).

We use the distractor benchmark of Trappolini et al. [13], covering BioASQ, Natural Questions, and PopQA. Each context contains a gold document flanked by two hard distractors; evaluation is restricted to samples answered correctly without distractors and incorrectly with them, yielding approximately 255 distractor-vulnerable samples per model. We compare against two logit-space baselines that correct the output distribution without altering internal representations: **Distractor-Contrastive Decoding (DCD)**, which subtracts a distractor-only logit component ($\mathscr{L}_{\text{DCD}} = (1+\alpha)\mathscr{L}_{GD} - \alpha\mathscr{L}_D$), and **Marginal Distractor Removal (MDR)**, which interpolates toward the gold-only distribution ($\mathscr{L}_{\text{MDR}} = (1-\alpha)\mathscr{L}_{GD} + \alpha\mathscr{L}_G$), evaluated at correction weight $\alpha \in \{0.5, 1.0\}$. Both require oracle knowledge of document roles and serve as depth probes: if output-layer corrections sufficed, representational interventions would be unnecessary.

3.2. Optimization and Evaluation

We optimize feature vectors with Adam ($\eta = 0.0005$) over the final W tokens of the input context at the specified SAE layers. Response fidelity is evaluated via a hierarchical LLM-as-a-judge protocol using Gemini 2.5 Flash [14] to filter null responses, verify named entities, and assess semantic equivalence. We quantify intervention efficacy using the **Trajectory Recovery Rate (ρ)**: the percentage of distractor-vulnerable instances for which the intervention produces a semantically correct response.

4. Results and Analysis

4.1. Recovery Performance

Table 1 reports the Trajectory Recovery Rate (ρ) across layer configurations and datasets. $L1$ interventions yield the highest recovery rates, reaching 71.35% (BioASQ) and 85.93% (NQ) for Llama-3.1. The superiority of $L1$ interventions supports the **causal suppression hypothesis**: deactivating distractor representations at early layers prevents their propagation through the residual stream. Gemma-2 exhibits greater depth stability, with $L4$ trailing $L1$ by only 2.26 points on NQ. PopQA exhibits a ceiling effect ($\rho > 90\%$) across both models. Window extension to W=400 saturates performance, indicating that layer selection dominates over window size. Our method substantially outperforms the logit-space

[2]Model checkpoints: Gemma-2-2B (https://huggingface.co/google/gemma-2-2b) and Llama-3.1-8B (https://huggingface.co/meta-llama/Llama-3.1-8B).

[3]SAE checkpoints: Gemma Scope (https://huggingface.co/google/gemma-scope-2b-pt-res) and Llama Scope (https://huggingface.co/OpenMOSS-Team/Llama3_1-8B-Base-LXR-8x/tree/main).

competitors: on NQ, $(L1, 400)$ exceeds the best alternative by 33.65 points for Gemma-2 and 42.57 points for Llama-3.1, confirming that distractor corruption cannot be corrected at the output layer alone.

Table 1
Trajectory Recovery Rate (%) comparison. Our method (bottom) is evaluated against the DCD and MDR competing methods (top). **Bold** indicates the best result per column. Configurations for our method are tuples (L, W), where L is the Intervention Layer and W is the Optimization Window.

	Gemma 2 2B			Llama 3.1 8B		
Config	**BioASQ**	**NQ**	**PopQA**	**BioASQ**	**NQ**	**PopQA**
Competing Methods						
$DCD_{\alpha=0.5}$	47.81	33.98	44.01	42.70	40.71	52.15
$DCD_{\alpha=1.0}$	53.78	44.66	56.34	44.94	44.69	64.11
$MDR_{\alpha=0.5}$	44.62	34.63	55.99	35.96	39.82	72.25
$MDR_{\alpha=1.0}$	48.21	46.93	75.35	37.08	43.36	78.47
Our Method						
$(L1, 10)$	66.93	75.40	91.55	66.29	84.79	**93.93**
$(L1, 100)$	**70.52**	76.05	92.61	70.79	85.17	93.12
$(L1, 400)$	**70.52**	**80.58**	93.31	71.35	**85.93**	93.52
$(L2, 10)$	69.32	77.99	92.61	66.85	81.37	91.50
$(L2, 100)$	68.13	76.38	92.25	65.17	82.89	92.71
$(L2, 400)$	66.53	79.94	92.61	64.61	83.65	92.31
$(L3, 10)$	68.13	78.32	92.96	71.35	83.27	91.90
$(L3, 100)$	68.13	79.61	92.96	70.79	85.55	91.09
$(L3, 400)$	68.13	78.32	91.55	**71.91**	82.51	90.28
$(L4, 10)$	68.92	78.32	92.96	67.98	81.75	89.88
$(L4, 100)$	68.13	77.67	93.31	66.85	79.85	89.47
$(L4, 400)$	68.92	78.32	**93.66**	64.61	80.61	89.88

Gemma 2 2B (26 layers): L1=0, L2=10, L3=18, L4=24. **Llama 3.1 8B** (32 layers): L1=0, L2=12, L3=22, L4=30.

5. Conclusions

This work characterizes the representational changes that occur when distractors induce generation errors. Our findings indicate that distractor interference operates through localized, correctable changes rather than global perturbations. Analysis of successful corrections reveals two complementary mechanisms: suppression of distractor features and amplification of discriminative gold-document cues. Lat-Defuse exploits this structure by identifying minimal modifications to a sparse set of interpretable features at the point where generation diverges from factual correctness. The correction mechanism generalizes across architectures (Gemma-2, Llama-3) and scales, achieving recovery rates up to 94%. By characterizing the representational basis of distractor interference, this work moves beyond symptom-level mitigation toward understanding root causes. Although currently dependent on ground-truth labels, these findings open directions toward universal distractor robustness in LLMs.

The intervention requires ground-truth supervision, which limits immediate deployment but establishes a foundation for future unsupervised methods.

Declaration on Generative AI

During the preparation of this work, the authors used Claude (Anthropic) for grammar and spelling check. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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